UE25CS243A-Computer Networks
Unit I- Question Bank

Chapter 1: Introduction

1) A university campus has three types of connections: student hostels using home-style broadband, computer labs connected within the university's own building network, and staff members accessing resources over their personal 4G phones while commuting. Classify each connection into the correct access-network category, and state one criterion that should be used to evaluate any access network. [4 Mark] [Easy]

2) An ISP's backbone link has a capacity of 2 Gbps. Each active user on this link requires 2 Mbps, and at any given instant only about 5% of all connected users are actively transmitting. Determine how many users circuit switching could guarantee on this link, and explain why packet switching can support substantially more users than this guaranteed figure. [4 Mark] [Medium]

3) A router uses store-and-forward packet switching to transmit packets of length L = 5000 bits over a link with transmission rate R = 50 Mbps. Calculate (a) the transmission delay for one packet, and (b) the maximum number of packets that can be transmitted onto this link in one second. [4 Mark] [Easy]

4) A single link connects two routers. The link length is d = 600 km, propagation speed s = 2×10^8 m/sec, packet length L = 6000 bits, and transmission rate R = 2 Mbps. Calculate the transmission delay, the propagation delay, and the total delay for this link. State which delay dominates, or whether they are comparable. [6 Mark] [Medium]

5) A router's output link has a constant transmission rate R = 2,000,000 bps and constant packet length L = 5000 bits. Calculate the traffic intensity and average queuing delay when the average packet arrival rate is (a) a = 300 packets/sec,

and (b) a = 350 packets/sec. Comment on the change in delay observed. [6 Mark] [Hard]

6) Explain why doubling the traffic intensity from I = 0.4 to I = 0.8 does not simply double the average queuing delay. [4 Mark] [Easy]

7) Consider two independent client-server download scenarios along a single bottleneck path: (a) Server sends at Rs = 8 Mbps, client link capacity Rc = 3 Mbps. (b) Server sends at Rs = 2 Mbps, client link capacity Rc = 6 Mbps. For each scenario, calculate the achievable end-to-end throughput and identify which link is the bottleneck. [6 Mark] [Easy]

8) Consider a caravan of 8 cars. A tollbooth services (transmits) one car every 3 seconds. Once serviced, cars travel at 60km/hr, and the next tollbooth is 30 km away. Assuming the entire caravan must be serviced before the last car begins traveling, find the total time from when the first car enters service until the last car arrives at the second tollbooth. [4 Mark] [Medium]

9) A user in Bengaluru finds that streaming from an origin server in the USA (without a CDN) gives a round trip of roughly 230ms, while streaming through a nearby CDN edge cache gives roughly 8ms. Explain why this improvement occurs. [4 Mark] [Easy]

10) Compare the repeater, hub, and switch in terms of the OSI layer at which each operates and the intelligence/scope of the forwarding decisions each can make. [4 Mark] [Easy]

11) A growing company's LAN which is built entirely on switches is beginning to suffer from broadcast storms as it scales. The company also wants all employees to share a single Internet connection securely, without exposing internal device addresses to the outside. Identify which device should be added to solve these problems and justify the choice by referencing its layer of operation and specific functions. [6 Mark] [Medium]

12) An ISP provides a 1 Mbps link to 50 users. [4 Mark] [Easy]

Each user requires 100 kbps only when active and is active 10% of the time.

(a) Explain the principle of statistical multiplexing.

(b) Why can packet switching support more users than circuit switching in this situation?

(c) What assumption makes statistical multiplexing effective?

13) A software company has 500 employees. [6 Mark] [Medium]

During office hours:

- Some attend video meetings.
- Others browse websites.
- Some download software updates.
- Many remain idle.

(a) Compare packet switching and circuit switching with respect to: Resource allocation, Delay, Scalability

(b) Which switching technique would you recommend?

(c) Justify your answer considering the company's traffic pattern.

14) A 16 KB packet travels from a client to a server through 4 communication links and 3 routers. [6 Mark] [Hard]

Each link has: Transmission rate = 20 Mbps, Propagation delay = 2 ms

Each router introduces: Processing delay = 10 µs, Queuing delay = Negligible

(a) Calculate the total transmission delay.

(b) Calculate the end-to-end delay.

(c) Which delay component contributes the most? Justify.

15) An ISP observes that every evening thousands of users start streaming 4K videos simultaneously. [4 Mark] [Easy]

(a) Explain how congestion develops in routers.

(b) How does congestion affect queuing delay and packet loss?

(c) Suggest two techniques that can reduce congestion.

16) A packet travels through the following communication path: [6 Mark] [Easy]

Link 1 = 1 Gbps

Link 2 = 250 Mbps

Link 3 = 100 Mbps

Link 4 = 500 Mbps

(a) Identify the bottleneck link.

(b) Determine the maximum achievable throughput.

(c) If only Link 4 is upgraded to 2 Gbps, will the throughput improve? Justify.

17) A 12 KB packet travels through 5 communication links. [4 Mark] [Medium]
Each link has: Transmission rate = 50 Mbps, Propagation delay = 1 ms
Each router introduces: Processing delay = 20 µs, Queuing delay = 0.5 ms
(a) Calculate the total transmission delay.
(b) Calculate the total end-to-end delay.
(c) If the queuing delay suddenly increases to 5 ms per router, explain how the overall network performance changes.

18) A university plans to expand its campus network. Different departments suggest using Twisted Pair Cable, Coaxial Cable, Optical Fiber, or Wireless Radio for various applications.
(a) Compare the characteristics of the four physical media.
(b) Identify one major advantage of each medium.
(c) Recommend the most suitable medium for a high-speed campus backbone network and justify your answer. [4 Mark] [Medium]

19) A packet switch receives packets at an average rate of a = 100 packets/second. Each packet is L = 8,000 bits long, and the outgoing link has a transmission rate of R = 1 Mbps. [6 Mark] [Hard]
(i) Compute the traffic intensity La/R.
(ii) Comment on whether the average queuing delay will be small, large, or infinite, and justify your answer.
(iii) If a increases to 125 packets/second, what happens to the queuing delay? Explain using the relevant formula/graph.

20) A college server needs to send a 12000-bit packet to a router over a link with transmission rate 10 Mbps before the router can forward it further. (a) How long does it take the router to receive the complete packet, and (b) what is the maximum number of such packets the link can carry per second? [6 Mark][Easy]

21) An office LAN administrator connects five computers to a single networking device so that they can communicate. Whenever anyone computer sends data, all the other four computers receive that same data, and only one computer can transmit at a time without collisions. Identify the device being used, explain why the described behavior occurs, and discuss one limitation of this arrangement compared to a more intelligent device. [4 Mark] [Easy]

22) A hospital's monitoring system sends small emergency-alert packets over a shared hospital network that also carries large routine file transfers. Explain, using store-and-forward switching, why a large file's packets don't block the network indefinitely for the alert packets. [4 Mark] [Easy]

23) A telecom company must lay a link between two cities 50 km apart, along a route that runs alongside high-voltage power lines and needs the maximum possible bandwidth with minimal signal degradation over the long term.
(i) Should guided or unguided media be used here, and why?
(ii) Which specific guided medium should be selected? Justify your choice.
(iii) Name one guided medium that would be a poor choice in this specific case and explain why. [6 Mark] [Hard]

24) A client downloads a 600 MB file from a server through three links with transmission rates of 80 Mbps, 25 Mbps, and 50 Mbps. Assuming there is no congestion and propagation delay is negligible, determine:

i. The end-to-end throughput.
ii. The time required to transfer the complete file. [4 Mark] [Easy]

25) A company operates two geographically separated data centers connected through multiple routers. During normal operation, users observe an average throughput of 180 Mbps. After upgrading all routers along the communication path, the throughput remains unchanged. Further investigation reveals that the sender's network interface card (NIC) supports only 200 Mbps transmission.
Answer the following:
a) Identify the factor limiting the end-to-end throughput. (2 Marks)
b) Explain why upgrading intermediate routers did not significantly improve the throughput. (2 Marks)
c) Suggest two network upgrades that could effectively improve the overall throughput and justify your answer. (2 Marks) [Easy]

26) Assume an application generates 2000 bytes of data. The following headers are added during transmission:
Transport Layer Header = 20 bytes
Network Layer Header = 20 bytes
Link Layer Header = 18 bytes
Link Layer Trailer = 4 bytes

Answer the following:

a) Determine the total frame size transmitted over the physical medium. (2 Marks)

b) Calculate the protocol overhead as a percentage of the original application data. (2 Marks)

c) Explain why protocol encapsulation is necessary despite introducing overhead. (2 Marks) [Easy]

27) A packet traverses two routers before reaching its destination. Each communication link has a transmission rate of 50 Mbps and a propagation delay of 2 ms. The packet length is 10,000 bits. Initially, the routers experience no queuing delay. Later, due to congestion, each router introduces an average queuing delay of 5 ms.

a) Calculate the end-to-end delay when there is no queuing delay. (3 Marks)
b) Calculate the end-to-end delay after congestion occurs. (2 Marks)
c) By what percentage has the total end-to-end delay increased because of queuing? (1 Mark) [Medium]

28) A message of size 4 MB is divided into 1000 equal-sized packets. These packets are transmitted from a source to a destination over three communication links, each having a transmission rate of 10 Mbps. Ignore propagation, processing, and queuing delays.

a) Determine the size of each packet in bits. (2 Marks)
b) Calculate the transmission delay for one packet on a single link. (2 Marks)
c) Explain why packet switching with pipelining completes transmission faster than sending the entire message as one large packet through store-and-forward switches. (2 Marks) [Medium]

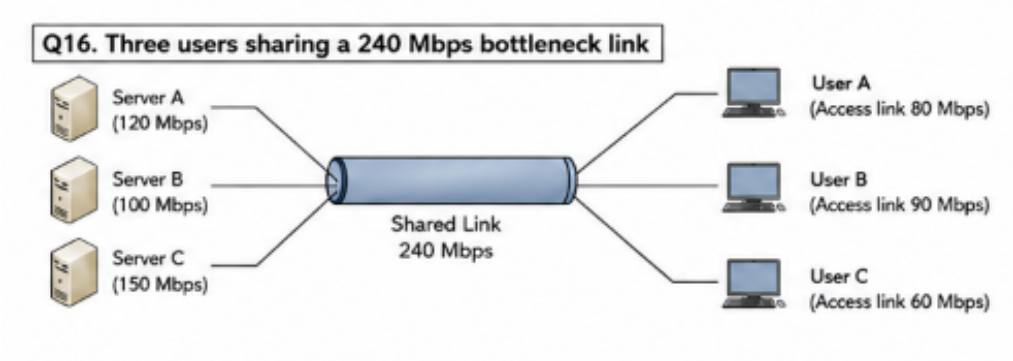
29) A newly established university is designing its complete campus network. The university requires:

- Internet connectivity for classrooms.
- High-speed communication between academic buildings.
- Wireless Internet for students.
- Reliable connectivity to the ISP.

(a) Identify the major components of the network edge required for this campus.

(b) Recommend suitable access networks and physical media for each requirement.
(c) Explain how these components work together to provide Internet connectivity to students and faculty. [6 Mark] [Hard]

30) Three users, A, B, and C, simultaneously download files from three different servers. Their communication paths share a common bottleneck link of capacity 240 Mbps. The individual access link capacities are shown below. [6 Mark] [Hard]



User	Server Access Link	Client Access Link
A	120 Mbps	80 Mbps
B	100 Mbps	90 Mbps
C	150 Mbps	60 Mbps

Assume the 240 Mbps bottleneck link is shared equally among the three users.

a) Calculate the maximum achievable throughput for each user. (3 Marks)
b) Identify the bottleneck link for each communication path. (2 Marks)
c) If User C upgrades the client access link to 120 Mbps, will the throughput improve? Justify your answer. (1 Mark)

Chapter 2: Application Layer

31) Two applications, a chat client and a web browser, are running simultaneously on the same laptop and both need to communicate over the Internet.
(a) Explain how a process on the laptop is identified so that incoming data reaches the correct application.
(b) What is a socket, and what role does it play in network communication?
(c) Why is a port number alone not sufficient to identify a process across the Internet? [4 Mark] [Easy]

32) A company is deploying four applications: file transfer, Internet telephony, video conferencing, and a stock trading application.
(a) Identify the key transport service requirements (data loss, throughput, timing) for each application.
(b) Recommend TCP or UDP for each application.
(c) Explain why TCP is not always the preferred choice, even though it is more reliable. [4 Mark] [Easy]

33) A user's browser requests a webpage containing an HTML file, 3 images, and 2 stylesheets from the same server.
(a) Differentiate between non-persistent and persistent HTTP connections.
(b) Explain how many TCP connections are required in each case to fetch all objects.
(c) Why do modern browsers and servers prefer persistent connections with pipelining? [6 Mark] [Medium]

34) An e-commerce website remembers a user's shopping cart items even after the user closes and reopens the browser tab.
(a) Explain how HTTP cookies enable this behavior, given that HTTP itself is stateless.
(b) Identify the four components involved in cookie-based state management.
(c) Discuss one privacy concern associated with the use of cookies. [4 Mark] [Easy]

35) A news website updates its homepage frequently, and a web cache stores a copy of it for faster delivery to users. [Medium]
(a) Explain the purpose of a conditional GET request.

(b) Describe the sequence of messages exchanged when the cached copy is still valid.
(c) What would happen if caches never checked for updated content? [6 Mark]

36) A company migrates its web servers from TLS 1.2 to TLS 1.3 and notices that HTTPS connections are established noticeably faster.
(a) Explain, at a conceptual level, what a TLS handshake accomplishes before data can be sent securely.
(b) Explain why TLS 1.3 typically requires fewer round trips to establish a secure connection than TLS 1.2.
(c) What does TLS 1.3's "0-RTT" (zero round-trip time) resumption feature allow for returning clients? [6 Mark] [Medium]

37) A webpage's base HTML file references 5 embedded images, all hosted on the same server. The round-trip time (RTT) between the client and server is 40 ms. Assume transmission delay of each object is negligible, and the browser opens connections sequentially (one at a time), with non-persistent, non-parallel HTTP connections (no pipelining, one object per new TCP connection).
(a) Calculate the time to establish one TCP connection (assume a standard TCP handshake).
(b) Calculate the total time to fetch the base HTML file plus all 5 images sequentially using non-persistent connections.
(c) How much time would be saved if the browser instead used one persistent HTTP connection for all 6 objects? [5 Mark] [Medium]

38) A user visits a website for the very first time (no DNS entries cached anywhere). The DNS resolution takes 2 RTTs to complete (RTT = 50 ms), and the HTTP object fetch uses a non-persistent connection. [6 Mark] [Medium]
(a) Calculate the total delay for DNS resolution.
(b) Calculate the additional delay to fetch the webpage's single HTML object after DNS resolves.
(c) Calculate the total page-load time, and explain how caching the DNS result would affect a second visit to the same site later.

39) A webpage has 1 base HTML file and 10 small embedded objects, all on the same server. RTT = 30 ms and transmission delays are negligible. [6 Mark] [Hard]
Compare three cases:

(i) non-persistent HTTP with sequential (one-at-a-time) connections,
(ii) persistent HTTP without pipelining, and
(iii) persistent HTTP with pipelining (all requests sent back-to-back without waiting for responses).

(a) Calculate the total page-load time for case (i).
(b) Calculate the total page-load time for case (ii).
(c) Calculate the total page-load time for case (iii), and comment on which is most efficient.

40) A single laptop is simultaneously running a web browser (making a request to an HTTP server) and an email client (checking mail from a mail server). Explain what identifier each process on this laptop needs so that response messages reach the correct application and give the two components that make up this identifier. [4 Mark] [Easy]

41) A software team is deciding whether to build a new interactive multiplayer game on top of TCP or UDP. They know the game needs frequent, fast position updates and can tolerate occasionally losing an update, but cannot tolerate the delay caused by waiting for retransmissions. Using the properties of TCP and UDP, recommend a transport protocol and justify the choice against alternatives. [6 Mark] [Hard]

42) A developer is building a simple e-mail client that must guarantee every message is delivered completely and in order, even if that means occasional delay. Which Internet transport protocol should this application use, and which specific TCP service guarantees make it suitable, referencing the guarantees TCP does NOT provide as well? [4 Mark] [Easy]

43) A client uses non-persistent HTTP (no parallel connections) to fetch a webpage consisting of 1 base HTML file and 8 embedded JPEG images, with each object requiring its own new TCP connection. The round-trip time (RTT) between client and server is 40 ms. The transmission time for the base HTML file is 2 ms, and the transmission time for each JPEG image is 3 ms. Calculate the total time to fully load the page. [4 Mark] [Medium]
Formula used:

- Non-persistent HTTP response time per object = 2·RTT + object transmission time

- Total page load time = Σ (response time for each of the 9 objects, fetched sequentially)

44) A user reports four different problems while browsing a site: [4 Mark] [Easy]
(a) a page they bookmarked a year ago now redirects somewhere else entirely
(b) a page shows a completely blank error saying the document could not be located
(c) a form submission returns an error saying the server could not understand the request, and
(d) the page they wanted actually loads fine and displays. Match each situation to the correct HTTP response status code and briefly justify each match.

45) A raw HTTP request captured on the wire begins with the line `GET /index.html HTTP/1.1`, followed by several header lines such as `Host:`, `User-Agent:`, and `Accept-Language:`, and ends with an empty line (just a carriage-return/line-feed). Explain what each of these three parts represents in the general HTTP request format, and what marks the boundary between the header lines and any message body. [4 Mark] [Easy]

46) A web application needs to:
(a) let a user submit a long comment through a form,
(b) allow a client to check whether a large file has changed without downloading the whole file,
(c) let a user search a site by typing keywords that appear directly in the browser's address bar,
(d) let a publishing tool completely replace an existing file on the server with a new version.
For each requirement, identify the correct HTTP method and briefly justify the choice. [4 Mark] [Easy]

47) An institutional network has an access link rate of 2 Mbps and a 1 Gbps internal LAN. The average request rate from browsers to origin servers is 10 requests/sec, and each requested Web object is 150 Kbits in size. Suppose a Web cache is installed, and it achieves a cache hit rate of 0.5 (50% of requests are satisfied at the cache, and 50% must go to the origin server over the access link). If the round-trip delay to origin servers (dominated by Internet delay) is 1.5 seconds and the cache-served delay is negligible, calculate

(a) the access-link utilization without a cache,
(b) the access-link utilization with the cache installed,
(c) the resulting average end-to-end delay with the cache installed. [6 Mark] [Medium]

48) HTTPS is described as "HTTP with encryption." Identify the three core security properties that the underlying TLS protocol provides to HTTPS traffic, and briefly explain the mechanism behind each. [4 Mark] [Easy]

49) A returning visitor's browser uses TLS 1.3's 0-RTT resumption mode to reconnect to a server, sending an HTTP GET request for their account page inside the very first packet of the new connection. Explain how this 0-RTT mode works, and evaluate whether it would be safe to use this same 0-RTT mechanism for a request that transfers money out of the user's account. [6 Mark] [Medium]

50) A browser connects to a server claiming to be "yourbank.com." Walk through, step by step, how TLS authentication allows the browser to confirm it is genuinely talking to yourbank.com's real server and not an attacker who has somehow obtained a copy of yourbank.com's public certificate. [4 Mark] [Medium]

51) A browser needs to establish a secure HTTPS connection and then send a single HTTP GET request. The RTT between client and server is 60 ms, and the object transmission time once the request is sent is 10 ms. Assume the total time = (RTTs needed for the TLS handshake) + (1 RTT for the HTTP request/response exchange) + (object transmission time). Compute the total time to load the page using (a) a TLS handshake requiring 2 RTTs, and (b) a TLS handshake requiring 1 RTT. State the time saved. [4 Mark] [Medium]
Formula used: Total secure page load time = (handshake RTTs × RTT) + (1 RTT for HTTP request/response) + object transmission time

52) List any four transport-layer service requirements that a network application may need [4 Marks] [Easy]

53) Identify which transport protocol — TCP or UDP — is more appropriate for this game, and justify your choice in 2–3 lines based on the timing and loss-tolerance needs described. [4 Marks] [Easy]

54) An institution's 1.54 Mbps access link is 97% utilized, causing large end-to-end delays. Two options are proposed — (a) upgrade the access link to 154 Mbps, or (b) install a local Web cache (hit rate 0.4). Given RTT to the origin server = 2 sec, object size = 100 Kbits, and request rate = 15/sec, evaluate both options and recommend one with justification. [7 Mark] [Hard]

55) A web browser on Host A requests a webpage from a web server running on Host B. The browser process uses source port 51000, while the web server listens on port 80. [6 Mark] [Easy]

a) Explain how the operating system identifies the correct destination process when the packet reaches Host B. (2 Marks)

b) Differentiate between the roles of the IP address and the port number in this communication. (2 Marks)

c) Suppose another application on Host B is listening on port 8080. Will the incoming HTTP request be delivered to that application? Justify your answer. (2 Marks)

56) A university plans to deploy the following campus applications: [6 Mark] [Hard]

- Learning Management System (LMS)
- Live online classroom
- Digital library for downloading lecture notes

a) Recommend the most suitable application architecture (Client–Server or Peer-to-Peer) for each service. Justify your answer. (3 Marks)

b)During the semester break, the Digital Library experiences a tenfold increase in simultaneous download requests. Explain why a pure Client–Server architecture may become a bottleneck and discuss how a Peer-to-Peer architecture can improve scalability in this scenario. (2 Marks)

c) The university requires centralized authentication and access control for all three services. Which architecture is more suitable for meeting this requirement? Justify your answer. (1 Mark)

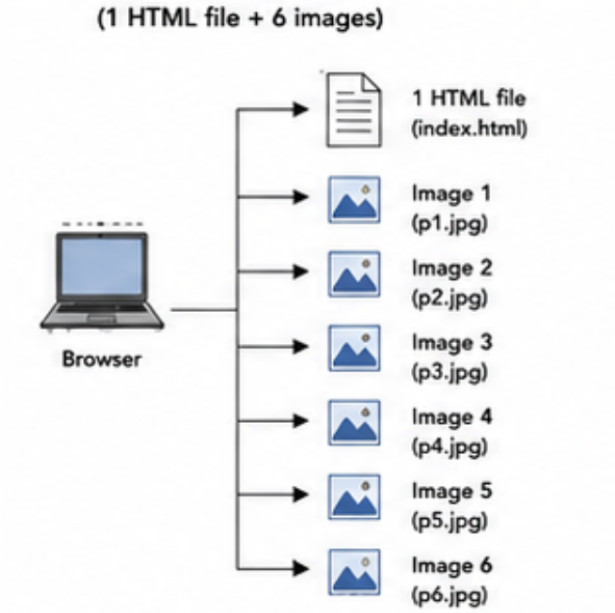
57) A laptop is simultaneously performing the following operations: [6 Mark] [Medium]

- Downloading a file from a cloud server
- Streaming an online lecture

- Browsing a university website

a) Explain how the operating system ensures that incoming data is delivered to the correct application process. (2 Marks)
b) Differentiate the roles of an IP address and a port number in this scenario. (2 Marks)
c) If the web browser opens five tabs connected to the same web server, explain how the operating system distinguishes between these connections. (2 Marks)

58) A webpage consists of: [6 Mark] [Hard]
1 HTML file
6 images
Assume:
One RTT = 80 ms
Object transmission time is negligible.



a) Calculate the total time required to download the webpage using Non-Persistent HTTP. (3 Marks)
b) Calculate the total time required using Persistent HTTP (without pipelining). (2 Marks)
c) Which approach provides better performance? Justify. (1 Mark)

59) A client sends the following HTTP request: [6 Mark] [Easy]

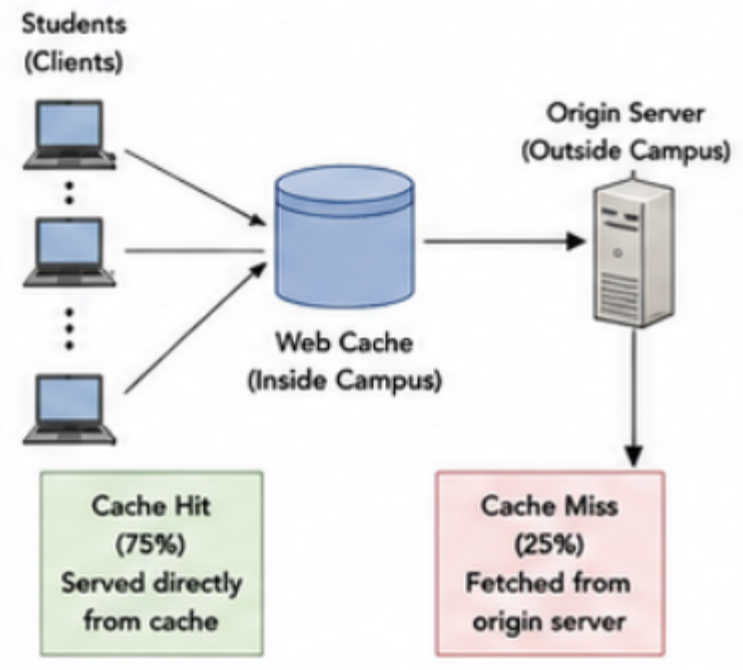
```
GET /courses/index.html HTTP/1.1
Host: pes.edu
Connection: keep-alive
User-Agent: Mozilla/5.0
Accept: text/html
Accept-Language: en-US
```

a) Identify the Request Line and explain each of its three fields. (3 Marks)
b) Explain the purpose of the Host and Connection header fields. (2 Marks)
c) State the HTTP status code expected in the server response. (1 Mark)

60) A web page consists of: [6 Mark] [Hard]
1 HTML page
8 embedded images
2 CSS files
Assume:
RTT = 40 ms
Transmission time for each object is negligible.
The browser opens 4 parallel TCP connections while using Non-Persistent HTTP.
a) Calculate the total time required to retrieve all objects using Non-Persistent HTTP with 4 parallel connections. (3 Marks)
b) Calculate the total time required if Persistent HTTP (without pipelining) is used. (2 Marks)
c) Which approach performs better in this scenario? Justify. (1 Mark)

61) A college campus has 600 students accessing the same online programming tutorial every morning. [6 Mark] [Easy]

- Each tutorial video is 80 MB.
- Without caching, every request is forwarded to the origin server.
- The college installs a web cache with a cache hit rate of 75%.



a) Calculate how many requests are served directly by the cache if 600 requests are received. (2 Marks)
b) Calculate how many requests are forwarded to the origin server. (2 Marks)
c) Apart from reducing Internet bandwidth usage, state two additional advantages of deploying a web cache. (2 Marks)

62) A university launches two websites: [6 Mark] [Easy]

Website A: Displays campus news.

Website B: Allows students to pay examination fees online.

a) Identify which website should use HTTPS and justify your answer. (2 Marks)

b) State two security services provided by TLS 1.3 that are essential for Website B. (2 Marks)

c) A student claims, "HTTPS makes websites faster than HTTP." Is this statement correct? Justify your answer.(2 Marks)

63) A student logs into the university ERP system using a web browser. After successful authentication, the student visits several different pages without entering the password again. [6 Mark] [Easy]

a) Explain how cookies enable the server to recognize the student during the browsing session. (2 Marks)

b) Describe the sequence of interactions between the browser, server and cookie database when a cookie is created and subsequently used. (3 Marks)

c) Why is a cookie ID stored in the browser instead of storing the complete user information? (1 Mark)

64) A college library has deployed a web cache. Initially, 120 students simultaneously request the same online lecture video. The cache is empty when the first request arrives. [6 Mark] [Easy]

a) Explain how the first request is processed by the web cache. (2 Marks)

b) Explain how the processing differs for the remaining 119 requests if the requested object is still valid in the cache. (2 Marks)

c) Discuss two advantages of using a web cache in this scenario. (2 Marks)

65) An e-commerce website wants to (a) recognize a returning user on its own site and (b) work with an ad network to show that user targeted ads on unrelated third-party websites. Explain, with a worked example, how first-party and third-party cookies achieve this, and discuss the privacy implication. [7 Mark] [Easy]

66) Explain, step by step, the TLS 1.3 handshake (1-RTT) between a client and a server. Also explain the 0-RTT resumption mode, its benefit, and its associated security risk. [8 Mark] [Hard]